

When Cross-View Helps Most: Conflict-Aware Disaster Triage Across Spatial Observation Regimes

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ABSTRACT

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage, but aggregate F1 hides examples where the two views disagree. This short paper introduces Conflict-Aware Evaluation (CAE), a reporting protocol that measures overall F1, conflict-subset F1, cross-view gain over the stronger single-view baseline, and the direction in which one view corrects the other. We apply CAE to an Eaton/Altadena wildfire dataset with property-centered ground imagery and to the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. Results differ by observation regime: on the wildfire sensitivity split, cross-view fusion improves conflict F1 by 0.0388; on the leakage-controlled hurricane endpoint split, the gain becomes negative ($\Delta_{\text{view}} = -0.0153$), although aggregate F1 makes cross-view and street-only models look nearly tied. Directional and alignment diagnostics show that fusion helps when the ground view observes the target building, but can lose value when the ground view becomes panoramic context. CAE therefore turns paired-view triage into a stress test for when fusion helps, fails, or should defer.

KEYWORDS

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

1 INTRODUCTION

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, but average test scores hide the cases that matter most for deployment: buildings where the views disagree.

This paper treats paired-view triage as conflict resolution. Each example has a ground image, an overhead patch, and a binary damage label. We ask whether fusion improves the subset where the single-view predictors disagree, rather than whether it only improves average F1. This framing matters because a property inspection photograph and a panoramic street scene are both ground views, yet they observe the labeled building in different ways.

The work builds on overhead damage benchmarks [2], fusion models for disaster assessment [5], ground-view damage studies [6–8], and the paired Hurricane Ian CVDIsaster benchmark [4]. CAE is related to selective prediction [1] because both settings identify cases where an automatic decision may need extra scrutiny,

Table 1: Evaluation settings, sample counts, and leakage controls.

Setting	Positive class	Ground regime	N	Role
WF sens.	affected+	property	19,720	main CAE
H end.	severe	panoramic	2,583	diagnostic
H mod.+sev.	moderate+	panoramic	4,121	sensitivity
H clean	severe	panoramic	2,583	leakage control

but CAE does not estimate risk coverage or abstention. Its reporting unit is spatial: the disagreement subset, the single-view correction direction, and the gain or loss from fusion under a given observation regime. The paper’s central finding is that spatial observation regime, meaning how a ground view aligns with the labeled building, is a stronger predictor of fusion value than aggregate test-set F1. CAE makes this finding measurable.

2 DATA AND PROTOCOL

2.1 Paired-View Triage Settings

Let $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$ be a paired disaster triage dataset with a ground image, an overhead patch, and a binary label. We compare street-only (S), remote-only (R), and cross-view (CV) predictors. CV uses late fusion of ResNet18 features [3]: both embeddings, their absolute difference, and their elementwise product.

Table 1 summarizes the four evaluation settings. The wildfire setting uses 19,720 usable Eaton/Altadena paired records derived from property-level inspection photographs and matched overhead crops. The source index retains object identifiers, coordinates, crop boxes, overhead tile IDs, and coverage metadata, enabling object-level grouping and tile-level spatial aggregation. The positive class is Affected + Minor + Major + Destroyed. The hurricane setting uses the Hurricane Ian CVDIsaster benchmark [4], with 4,121 paired samples across minor, moderate, and severe damage. We evaluate both the endpoint task (minor versus severe) and a broader moderate-plus-severe task. We use these settings to contrast property-centric and panoramic observation regimes.

2.2 Conflict-Aware Evaluation

For each example, let $p_i^{(s)}$ and $p_i^{(r)}$ denote the positive-class probabilities predicted by the street-only and remote-only models. CAE defines the conflict subset

$$C_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with $\tau = 0.1$ in the main analysis. For each model m , CAE reports standard overall F1 and conflict F1 on C_τ . It then computes the

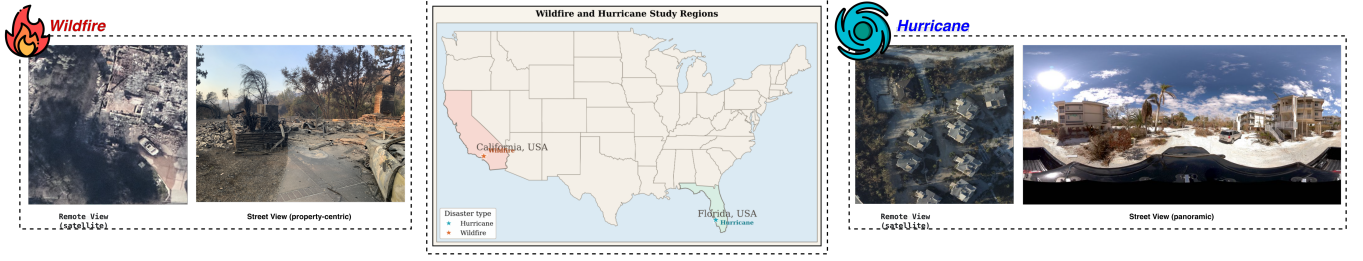


Figure 1: Study design and spatial observation regimes. The wildfire setting uses property-centered inspection imagery paired with overhead patches, whereas the hurricane setting uses panoramic ground imagery paired with overhead patches. This visual difference motivates the regime-aware interpretation tested by CAE.

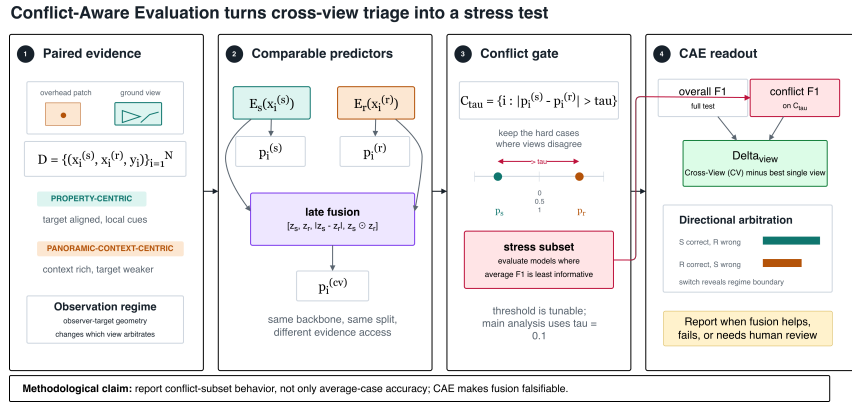


Figure 2: Conflict-Aware Evaluation (CAE) protocol. CAE isolates examples where street-only and remote-only predictors disagree, then reports whether cross-view fusion improves conflict-subset F1 over the stronger single-view baseline. Directional arbitration links model behavior to spatial observation regime.

cross-view conflict gain

$$\Delta_{\text{view}} = F1(\text{crossview}; C_{\tau}) - \max\{F1(\text{street}; C_{\tau}), F1(\text{remote}; C_{\tau})\}. \quad (2)$$

The compact CAE report for each model is therefore

$$\mathbf{r}_{\tau}(m) = (F1(m; \mathcal{D}), F1(m; C_{\tau}), |C_{\tau}|/N). \quad (3)$$

We also decompose conflicts by correction direction. Let $\hat{y}_i^{(s)} = 1\{p_i^{(s)} \geq 0.5\}$ and define $\hat{y}_i^{(r)}$ analogously. With $c_i^{(v)} = 1\{\hat{y}_i^{(v)} = y_i\}$, the two arbitration rates are

$$\rho_{S \rightarrow O} = \frac{1}{|C_{\tau}|} \sum_{i \in C_{\tau}} c_i^{(s)} (1 - c_i^{(r)}), \quad (4)$$

$$\rho_{O \rightarrow S} = \frac{1}{|C_{\tau}|} \sum_{i \in C_{\tau}} c_i^{(r)} (1 - c_i^{(s)}). \quad (5)$$

$S \rightarrow O$ denotes street-correct and remote-wrong cases, and $O \rightarrow S$ denotes the reverse. The original hurricane endpoint split allowed overlapping object identifiers across train and test through panoramic images tied to multiple labeled footprints. We therefore re-split with object-level grouping so no identifier appears in both partitions. The grouped-clean endpoint split is the conservative control.

2.3 Implementation Details

Images are resized to 224 by 224 pixels and normalized for ImageNet-pretrained ResNet18 encoders. Each view embedding is projected to 256 dimensions. Models use AdamW with learning rate 10^{-4} for 10 epochs. Grouped splits use an 80/10/10 train/validation/test design with approximate label stratification, and the best-validation-F1 checkpoint is evaluated on the held-out test split. Stability checks use seeds 42, 123, and 456. Cross-view validation F1 is 0.9689 ± 0.0011 on wildfire and 0.9455 ± 0.0063 on hurricane; street-only baselines are close but lower (0.9657 ± 0.0001 and 0.9433 ± 0.0012).

For reproducibility, the wildfire release is organized around split manifests and a data-card summary rather than raw-image redistribution. The manifests keep object IDs, image-source fields, coordinates, matched overhead tile names, binary labels, and filtering flags, making the grouped split auditable when future work reconstructs the benchmark from the original imagery sources.

Backbone checks rule out a small-CNN artifact. On cross-view validation, ResNet18/ResNet50/DINOv2/CLIP obtain 0.9676/0.9693/0.9378/0.7502 on wildfire and 0.9694/0.9470/0.7975/0.6437 on hurricane. The supervised CNN features remain strongest, suggesting that generic pretraining transfers imperfectly to this domain-specific binary damage task.

Table 2: Conflict-Aware Evaluation summary. Conflict F1 uses C_τ with $\tau = 0.1$. WF is wildfire; H is hurricane. Gain is reported only for cross-view rows.

Setting	M	All	C-F1	Gain	$ C_\tau $
WF sens.	S	0.9392	0.6230	–	435
WF sens.	R	0.9238	0.5237	–	435
WF sens.	CV	0.9473	0.6618	+0.0388	435
H end.	S	0.9055	0.6842	–	32
H end.	R	0.8969	0.5806	–	32
H end.	CV	0.9192	0.7647	+0.0805	32
H mod.+sev.	S	0.8738	0.7324	–	128
H mod.+sev.	R	0.8873	0.7742	–	128
H mod.+sev.	CV	0.8915	0.7867	+0.0125	128
H clean	S	0.9016	0.8101	–	80
H clean	R	0.8385	0.6526	–	80
H clean	CV	0.9008	0.7949	-0.0153	80

3 RESULTS

Table 2 is the CAE scorecard. Average F1 alone would make CV look mostly helpful or tied. Conflict F1 changes the reading. Wildfire has a clear positive conflict gain (+0.0388), while the leakage-controlled hurricane split has a negative gain (-0.0153) even though S and CV are nearly tied overall. The hurricane conflict subsets are small ($|C_\tau| = 32$ and 80 for the endpoint settings), so their direction is a boundary signal to replicate rather than a high-precision effect estimate.

Figure 3 explains why the mixed gains are useful rather than inconvenient. In wildfire, $S \rightarrow O$ conflicts are more common than $O \rightarrow S$ conflicts (0.271 versus 0.189), so property-centered ground imagery is the stronger arbitrator. In the broader hurricane split, the ordering reverses (0.109 versus 0.133), which is consistent with panoramic views that contain more scene context and weaker target alignment. The endpoint hurricane split is closer to balanced (0.375 versus 0.344), reinforcing that panoramic regimes are split-sensitive.

Additional checks support the same ordering without changing the claim. At $\tau = \{0.1, 0.3, 0.5\}$, CV remains strongest on wildfire conflict accuracy (0.8349, 0.7821, 0.7538). At $\tau = 0.1$, wildfire conflict accuracy is 0.7885 for CV, 0.7356 for S, and 0.6529 for R; the hurricane moderate-plus-severe split gives 0.7500, 0.7031, and 0.7266. As τ increases, only stronger single-view contradictions remain, so accuracy can drop while the wildfire ordering persists.

4 DISCUSSION

The most important result in Table 2 is the negative grouped-clean hurricane gain: more visual evidence does not always improve decisions once object-level leakage is controlled. CAE therefore exposes a deployment boundary, not only a performance gain.

The conflict subset is not meant to replace the full test set. It is a second readout that asks a narrower deployment question: when two views support different decisions, does fusion add evidence or blur the better view? Unlike loss-based hard-case selection, the gate uses only single-view probabilities and can be computed before trusting fusion. Directional counts then identify whether the street view or overhead view supplies the missing cue.

This distinction matters for method comparison. Overall F1 describes the average case, whereas conflict F1 asks whether fusion

helps where the pipeline is least certain. A fused model can be useful for broad screening yet unsafe for escalation if it loses to a single view on C_τ . CAE keeps these interpretations separate.

Spatial observation regime makes this boundary plausible. In the wildfire records, the photographer usually stands near the parcel and aims at the structure that receives the label. The ground image can therefore carry direct evidence about charred walls, damaged openings, debris against the facade, or an intact exterior. In the hurricane records, the panoramic image often captures the street segment rather than a single labeled footprint. It may show flooded roads, vegetation loss, roof debris, and neighboring buildings, but the target building can be small, peripheral, or occluded. The mechanism is therefore how each view observes the target, not simply “street versus satellite.” The same architecture sees two views in both settings, but the semantic access of each view differs.

The tile-level panel adds a geographic check on this interpretation. Wildfire damage, disagreement density, and cross-view gain are spatially uneven across overhead tiles. High damage rate does not automatically imply high conflict, and high conflict does not automatically imply positive CV gain. This matters because deployment errors are rarely uniform: a model can look strong in a pooled metric while failing in parcels with unusual viewing angle, smoke shadow, vegetation cover, or partial structure visibility. CAE therefore works as both a model-comparison metric and a spatial audit layer.

Operationally, cases where S and R agree may use the agreement decision; cases where they disagree should route to CV or human review. Future datasets should collect property-centered and panoramic views for the same target buildings to test target alignment directly. The threshold τ is an operating knob: a small value measures broad disagreement, while a larger value keeps only the strongest contradictions when human review capacity is limited. The main analysis uses $\tau = 0.1$ because it leaves enough examples for stable reporting while still isolating visible disagreement.

Negative Δ_{view} is therefore informative. It says that the fused representation did not preserve the stronger single-view evidence under the current observation regime. In response use, this suggests routing: use automatic decisions on agreement cases, route high-disagreement cases to CV only when validation shows positive conflict gain, and otherwise defer to the dominant view or human review.

The study still has limits. The wildfire and hurricane settings differ in hazard type, geography, collection protocol, and label construction, so the paper treats observation regime as an explanatory lens rather than a causal proof. The compact image-alignment proxies in Figure 3 are diagnostic, not a replacement for field-verified viewpoint geometry. A stronger benchmark would collect several ground-view regimes for the same buildings and report CAE under identical spatial splits.

Two implementation details also deserve care in future work. First, the fusion model is simple by design to isolate the evaluation question from architectural choice, so the negative grouped-clean hurricane result should not be read as a universal limit of cross-view learning. It shows that late fusion can fail to improve the hardest cases after leakage control. Second, object grouping matters as much as architecture because visually similar views of the same object across splits can mimic cross-view reasoning. CAE

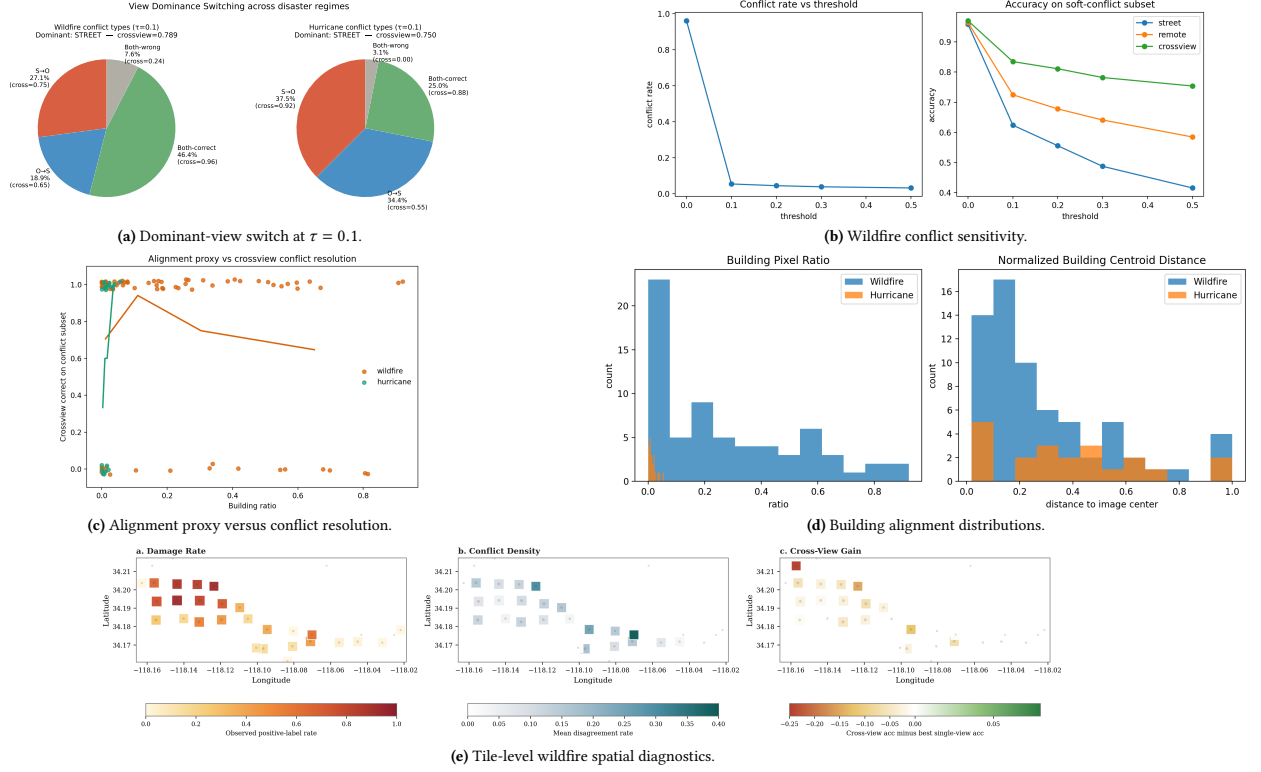


Figure 3: Result diagnostics. (a) Directional conflicts show the dominant-view switch. (b) Wildfire CV remains strongest across thresholds. (c–d) Alignment diagnostics support the regime interpretation. (e) Tile-level maps localize where fusion helps or fails.

is most useful when paired with grouped splits, transparent manifests, and geographic diagnostics.

5 CONCLUSION

CAE reframes paired-view disaster triage as a conflict-resolution problem. It shows a positive wildfire conflict gain, a negative grouped-clean hurricane gain, and a dominant-view switch tied to spatial observation regime. Disaster benchmarks should therefore report conflict-subset behavior, directional arbitration, and threshold sensitivity alongside aggregate scores.

The compact reporting template is $r_\tau(m)$ for each view and the fused model, Δ_{view} , the absolute conflict count $|C_\tau|$, and the $S \rightarrow O$ versus $O \rightarrow S$ correction balance. Paired with grouped splits and a small spatial summary, this template turns both positive and negative fusion results into useful boundary conditions. A stronger future model should not only improve pooled F1, but also reduce the unresolved conflict left by its single-view components.

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